



NIKE SALES ANALYSIS

Analyzing the sales data for NIKE to provide descriptive and predictive information about the underlying market trends, performance and growth as well as future scope.



JANUARY 26, 2025
MADE BY: - KRISH VISHNOI

DESCRIPTION

1. This is a self-made project with the aim of understanding the work of a Data/Business Analyst or any likewise job profile, I did it undermining each and every professional procedure to get a complete hand-on.
2. The project is based on the NIKE SALES data present and taken from Kaggle, it involves multiple parameters that could and have been undertaken for analysis, I used this dataset because it was messy and raw and needed to be transformed, which eventually helped me practicing my pre-processing and data cleaning skills.
3. My aim is to provide insights, trends, patterns, and performance hidden in the data to help the business understand the market and move in likewise direction.
4. I used python and a number of libraries (Numpy, Pandas, datetime, etc) for cleaning and preprocessing the data.
5. This project involves a POWER-BI dashboard with interactive visual helping the stakeholders understand data easily and in-depth as much it is possible.
6. I used to simultaneously upload it on my github rather than firstly completing it and then uploading the final version, this helped me understanding the usefulness, importance for github and git and also gave me hands-on experience using the same.

DETAILS FROM THE DATASET

1. ATTRIBUTES IN DATASET:

- a. Order_ID
- b. Gender_Category
- c. Product_Line
- d. Product_Name
- e. Size
- f. Units_Sold
- g. MRP
- h. Revenue
- i. Order_Date
- j. Sales_Channel
- k. Region
- l. Profit

2. ATTRIBUTES INFORMATION:

a. Order_ID	2500 non-null	int64
b. Gender_Category	2500 non-null	object
c. Product_Line	2500 non-null	object
d. Product_Name	2500 non-null	object
e. Size	1990 non-null	object
f. Units_Sold	1265 non-null	float64
g. MRP	1246 non-null	float64
h. Discount_Applied	832 non-null	float64
i. Revenue	2500 non-null	float64
j. Order_Date	1884 non-null	object
k. Sales_Channel	2500 non-null	object
l. Region	2500 non-null	object
m. Profit	2500 non-null	float64

CLEANING & PRE-PROCESSING

1. CLEANING AND PRE-PROCESSING SIZE COLUMN:

a. Problems:

- i. The **Size** column had mixed entries in it, such as L, XL, M, 6,7, 8, 9, 10, 11, 12.
- ii. The datatype was **string/object**.
- iii. There were 510 null values.

b. Solutions:

- i. Due to mixed and alphabetical entries data quality was at risk, hence the alphabetical entries were firstly converted into their respective numerical representative in the **string** format to match the other numerical entries as they were also in **string/object** format. The conversion was such that **M -> 8, L -> 10, XL -> 12**. The exact conversion numbers were decided gathering the domain knowledge based on Indian region using google and visiting some ecommerce application to be sure.
- ii. Once the alphabetical entries were converted into their respective numerical representative, python merged the same one automatically with the other equal and same **string**, after that when all the entries were numerical but in string format, the **Size** column was then converted to **int64** datatype format.
- iii. To fill the null values based on the information available in the dataset itself, it was decided to fill the values with the median based on the occurrence in combination of their respective "**Gender_Category**", "**Product_Line**", because these were the most suitable attributes and information present in the dataset to represent the size alongside them. But before making it final decision it was checked that whether we have enough information and entries per group of "**Gender_Category**" & "**Product_Line**" and also it was checked that what unique values are present as per the group, after getting enough support and acceptable amount of information from within the dataset, the null values were filled with the median of their respective group.
for example: The null values in size column against "**Gender_Category = kids**", "**Product_Line=basketball**" were filled with the median of the sizes when "**Gender_Category = kids**", "**Product_Line = basketball**".

2. CLEANING AND PRE-PROCESSING Units_Sold COLUMN:

a. Problems:

- i. There were 205 values which were -1.
- ii. There were 224 values which were 0.0.
- iii. There were 1235 null values in the column.

b. Solutions:

- i. The values which were -1 were decided to be replaced by the positive value as the dataset didn't contain any return, hence all the -1 was replaced with +1.
- ii. The values which were 0.0 were definitely the erroneous values for the analysis hence were replaced with the median of non-zero value of their respective group category.
- iii. Once the negative values and zero values were handled, the null values were filled with median of their group-wise representative which were having the information. Before finalizing the median parameter, it was compared with mean and mode as well, the best and most acceptable reliable outputs were seen on median only. Same as **Size** column, the **Units_Sold** column was also grouped based on "**Gender_Category**" & "**Product_Line**" to make the fillings more meaningful and derived from the patterns and trends present in dataset itself.

3. CLEANING AND PRE-PROCESSING Order_Date COLUMN:

a. Problems:

- i. There were 616 null values in **Order_Date** column.
- ii. The dataset contained mixed data format in **Order_Date** column.
- iii. The column's datatype was object instead of **datetime**.

b. Solutions:

- i. The dataset was firstly sorted based on the **Order_Date** column, because It was decided that the null values will be filled using the forward-fill method because there was no other evident options or information present in the dataset for this task. The sorted dataset was then used to fill the null values using the forward-fill method and afterwards it was sorted based on **Order_ID**.
- ii. There were mixed date formats in the **Order_Date** column, leading to problems while changing the column's data type into datetime, hence all the date were converted into a common datetime format i.e. DD/MM/YYYY.
- iii. The column's datatype was then converted to datetime to match the values stored inside the column.

4. CLEANING & PRE-PROCESSING THE **Discount_Applied** Column:

a. Problems:

- i. The **Discount_Applied** column had 1668 null values.
- ii. During further inquiry it was found that the column was holding percentage of the discount but not the absolute value in currency that was subtracted from the revenue.
- iii. During further inquiry it was found that the **Discount_Applied** column is actually holding the percentage of the value paid to customer, it was found when the **Revenue** was compared with the actual process, and the calculation pointed that it is the percentage of amount actually paid not the discount applied on the revenue.

b. Solutions:

- i. The name of the column was changed to **Discount**.
The null values were filled with the group-wise median of their respective categories based on **Gender_Category & Product_Line**, eventually as there were a large number of null values compared to information below and because the groups were holding outlier values, it was decided to use median instead of mean because it was providing more accurate and acceptable figures for the discount.
- ii. The other two problems were solved and taken care of while handling the Revenue column.

5. CLEANING & PRE-PROCESSING THE **MRP** Column:

a. Problems:

- i. The only problem in **MRP** column was 1254 null values.

b. Solutions:

- i. The null values in **MRP** was filled with median of their group-wise category representative, the median was selected after comparing the diversity present in each group category which came to be present in almost every column and mean was giving almost identical **MRP** for every category which is practically not possible, hence median was considered.
To make the substitutions more concrete and reliable the grouping was done on three columns i.e. **Gender_Category, Product_Line, Product_Name** to make more in-depth decisions on the pattern in the group-wise information available.

6. CLEANING & PRE-PROCESSING THE Revenue Column:

a. Problems:

- i. Doing calculations and checking values in **Revenue** it became evident that **Discount** column is holding the percentage of amount paid by the consumer and this is an issue with the column name.
- ii. **Revenue** column had some 0.0 values but against that profit was mentioned.
- iii. The **Revenue** column even after several tries hold that showed 100 percent discount but yet profit was mentioned.

b. Solutions:

- i. As there was misinformation In the **Discount** column, two new columns were created named as **%_pricepaid** (this holds the exact same value in the **Discount** column but justifies the column name), a new **Discount%** column with newly and correctly calculated actual discount percentage. Finally, the misleading **Discount** column was dropped once we gathered and effectively represented the information stored in it.
- ii. The **Revenue** column was then re calculated for the values, it was verified that the values that were already present were calculated correctly and the values which were 0.0 or anything different, they were replaced with actual calculated values. A **Cost_Price** column was also created using the formula **Revenue – Profit** to add some details in the analysis and find out the hidden information about the business performance and numbers involved.
- iii. After the final check It was found that there are still some data rows that shows 100% discount yet have some profit against them, the issue could not be given a reason or solution, hence these rows were preserved in a separate variable and deleted from the main the dataframe.

7. CLEANING & PRE-PROCESSING THE Region Column:

a. Problems:

- i. The **Region** column had some mistyped value in it. Such as Hyd, Hyrderbad, etc.

b. Solutions:

- i. The mistyped or erroneous region names were replaced with the real and correct ones.

AT THE END ADDED THREE COLUMNS -> Quarter, Year, Month EXTRACTING THESE USING THE datetime LIBRARY IN PYTHON, THIS WAS DONE FOR MORE DATA REGARDING THE ANALYSIS.